

The training objective of RETRODIFF derives from the variational lower bound (ELBO) on the log-likelihood of clean data x . Applying Jensen’s inequality to $\log p_\theta(x)$ yields

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} | x)} \right] = -\mathcal{L}_{\text{ELBO}}, \quad (1)$$

where q is the fixed forward noising process that adds Gaussian noise over T steps, and p_θ is the learned reverse process. The bound is tight when $q(\cdot | x) = p_\theta(\cdot | x)$; in practice we minimise $\mathcal{L}_{\text{ELBO}}$ as a proxy for maximising $\log p_\theta(x)$.

Markov-decomposing $p_\theta(x_{0:T})$ and $q(x_{1:T} | x)$ step by step, and collecting terms by timestep, converts (1) into a reconstruction term plus a sum of per-step KL divergences:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0 | x_1)]}_{\text{reconstruction}} + \sum_{t=2}^T \underbrace{\text{KL}(q(x_{t-1} | x_t, x) \parallel p_\theta(x_{t-1} | x_t))}_{\text{step-}t \text{ denoising}}. \quad (2)$$

Each KL term penalises the reverse model $p_\theta(x_{t-1} | x_t)$ for deviating from the analytic posterior $q(x_{t-1} | x_t, x)$, which is Gaussian with closed-form mean when the forward process is linear-Gaussian. Equation (2) is standard for denoising diffusion probabilistic models; the only free parameters are the neural-network weights θ .

RETRODIFF’s contribution is to condition each denoising step on a set of k exemplars $R(x)$ retrieved from the training corpus given x . Averaging (2) over the data distribution with the retrieval conditioning in place gives the RETRODIFF objective:

$$\mathcal{L}_{\text{RETRODIFF}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}(x | R(x))]. \quad (3)$$

Here $R(x)$ denotes the k -nearest-neighbour exemplar set retrieved for sample x , and the ELBO in (3) is computed with $p_\theta(\cdot | \cdot, R(x))$ replacing the unconditional reverse process. Minimising (3) encourages the model to exploit retrieved structure while retaining the well-understood ELBO convergence guarantee of (1)–(2). $\square$